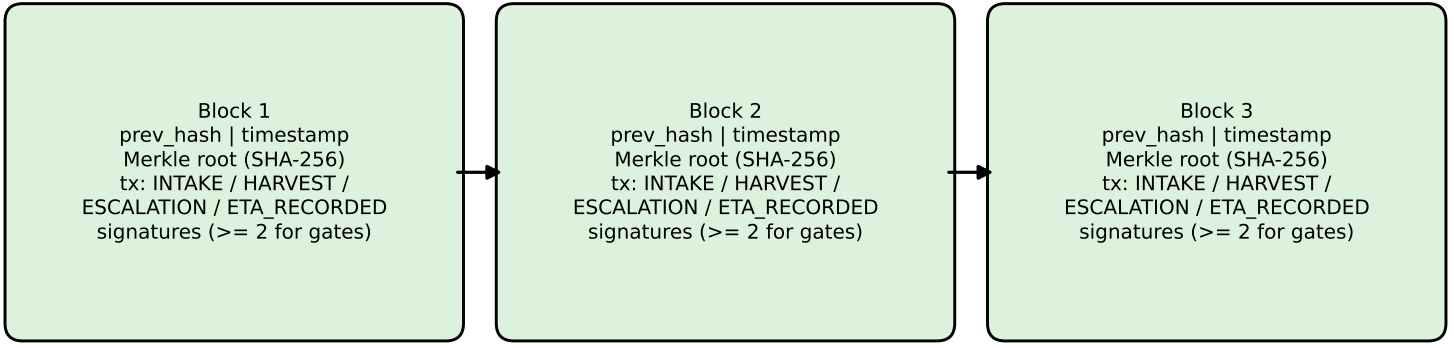Ledger structure: hash-linked blocks with Merkle-rooted transactions